

Data Estimation

Assignment 3

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1 Aim

1. Learning the intricacies of python functions
2. Learning the concept of partial application for a function
3. Curve fit
4. Working with approximating large and small numbers in python

2 Procedure

2.1 Reading data from the file

Read the data from the csv file.

2.2 Making the equation

The following equation for spectral radiance of a body is used to determine the best fit. Omit the differential terms, as they are not relevant for graph plotting.

$$B_{\lambda}(\lambda, T)d\lambda = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1} d\lambda,$$

Figure 1: The spectral radiance of a body.

2.3 Curve fitting with approximated initial values

Use the ‘curve_fit’ function from ‘scipy.optimize’ by providing the function and wavelength values as inputs. Initialize the parameters with values close to the expected target to achieve a reasonable approximation. This is important because the wide range of values makes it difficult to achieve convergence without proper initialization.

2.4 Curve fitting with partial application

2.4.1 Creating the function

Create a function that performs partial application by fixing the following constants:

Planck’s constant, $h = 6.626 \times 10^{-34}$,

Speed of light, $c = 3 \times 10^8$,

Boltzmann constant, $k = 1.3 \times 10^{-23}$,

and accepts wavelength (λ) and temperature as inputs to produce an output.

2.4.2 Applying curve fit

Use this newly created function in the ‘curve_fit’ function to obtain the best estimated value for the temperature.

3 The plots:

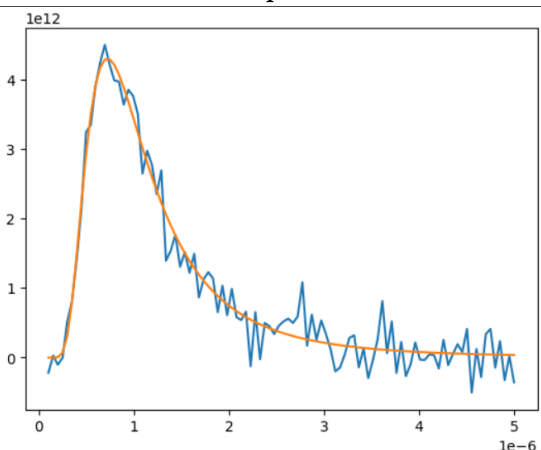
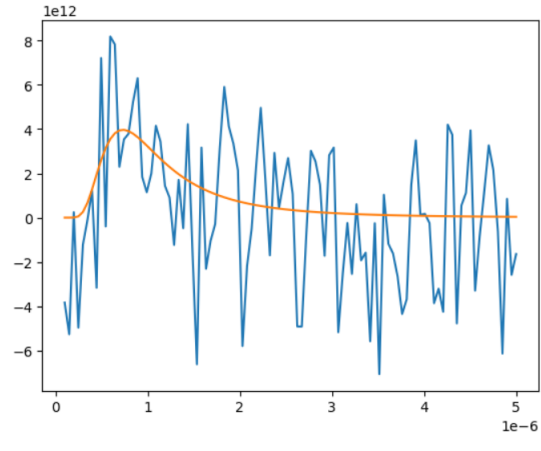
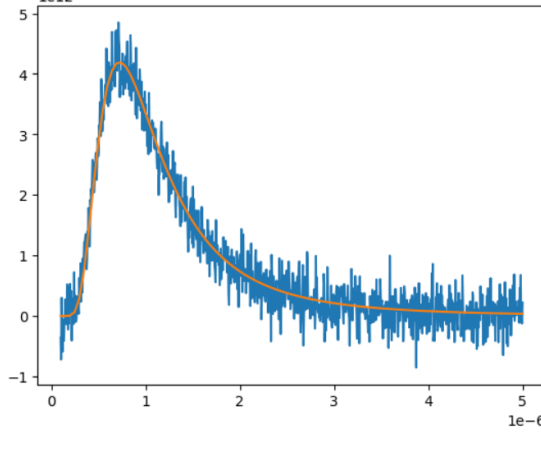
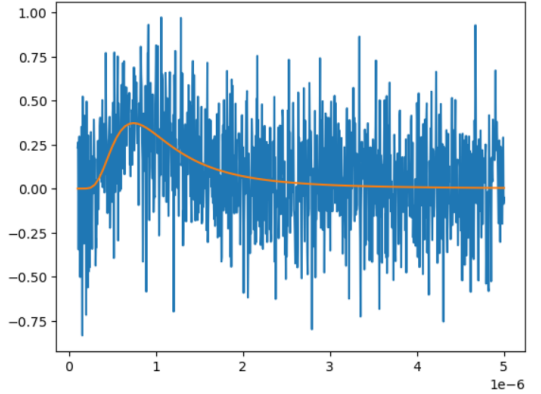
Category	plot
Data 1	
Data 2	
Data 3	
Data 4	

Table 1: Table with all the plots

4 Results

The partial application method provides more accurate estimates compared to the standard curve fitting, as evident from the lower squared error loss between the two approaches.

This improvement is particularly noticeable in the temperature estimates. In all four graphs, the temperature is consistently approximated to around 4000 K when using partial application.

5 Instructions for Running the IPYNB File

1. Update the File Paths:

In the notebook, replace the CSV file names with the full file paths to the corresponding `.csv` files on your system. Ensure the paths are correct and accessible by the notebook.

2. Execute the Script:

After updating the file paths, run all code cells sequentially to reproduce the results. The script will use the specified `.csv` files for the required operations.

3. Verify the Output:

Once all cells have executed, ensure the results are correctly reproduced. If any output files or logs are generated, verify their accuracy.

4. Important Notes:

- Make sure the directory where the CSV files are saved is writable.
- If errors occur, double-check that the file paths are correct and that the files are properly formatted as CSV.

Thank you